

Task 5 Sensor Data & Telemetry

Reference

Week 7 Portfolio Reference

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Target: 30-Record Demonstration

Dataset

PROJECT DOCUMENTATION & DATASET SPECIFICATION

Synthetic Demonstration Data Notice: The dataset referenced herein consists of exactly 30 synthetic demonstration records designed to simulate typical ambient room cycles, light-dependent threshold transitions, manual override testing, and analytical CSV exports. It does not represent uncalibrated physical measurements.

1. Dataset Schema & Field Descriptions

Field Name	Data Type	Typical Range	Logical Relationship / Description
Timestamp	ISO-8601 String	2026-09-29 18:00 to 18:58	2-minute interval sampling timestamps
Temperature_C	Float (1 decimal)	23.0°C to 32.0°C	Simulated ambient room temperature trend
Humidity_Percent	Float (1 decimal)	45.0% to 80.0%	Simulated relative humidity tracking
LDR_Value	Integer (12-bit)	580 to 2450	ADC raw value: > 1200 indicates BRIGHT, ≤ 1200 indicates DARK
Light_Status	Categorical String	BRIGHT DARK	Automated binary evaluation of the LDR voltage
Bulb_Status	Categorical String	ON OFF	Relay coil state (active HIGH driving AC bulb)
Control_Mode	Categorical String	AUTO MANUAL	Current operating firmware regime

2. State Transition & Consistency Validation

- Records 1–7 (Daylight):** LDR > 1200 (BRIGHT), Mode = AUTO → Bulb is OFF.
- Records 8–14 (Dusk):** LDR drops ≤ 1200 (DARK), Mode = AUTO → Relay activates, Bulb turns ON.
- Records 15–18 (Manual Override):** Mode = MANUAL → Bulb explicitly toggled OFF then ON regardless of darkness.
- Records 19–24 (Night Operation):** Mode = AUTO → Bulb resumes automatic illumination in dark conditions.
- Records 25–27 (Flashlight Test):** Sudden high illumination (LDR > 2100) → Mode = AUTO automatically extinguishes bulb.
- Records 28–30 (Recovery):** Flashlight removed (LDR < 800) → Automatic illumination restored.

3. Export Specifications

The dataset is stored in `task5_sensor_data.csv` and can be exported directly from the Task 5 interface via the **Export CSV** button. The file is encoded in UTF-8 with RFC 4180 standard CRLF line breaks.